Policy Modeling in Risk-driven Environment

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ABSTRACT

In this paper, the main challenges of applying ICT in policy modeling are described and a solution is proposed, which emphasizes policy impact exploration, monitoring and risk management. State of the art of policy modeling is given, with the summary of those features of our solution, which goes beyond the available approaches. The paper will be structured as follows: First, the challenges of the ICT utilization for policy modeling are detailed. Next theoretical background of policy modeling is discussed, followed by research overview. The proposed solution – policy modeling cycle and the corresponding system is presented in the following section. Finally, conclusion and future work are shown.

Categories and Subject Descriptors

J.1 [Administrative Data Processing]: Government

General Terms

Management, Design, Experimentation.

Keywords

e-Government, ICT, Policy modeling, Policy Impact, Risk management, Semantic Technology

1. INTRODUCTION

The popularity of the social media, and a variety of participatory tools such as web/text/opinion mining systems, online social networking, blogs, wikis, and forums, present government decision-makers, governance bodies and civil society actors with the possibility to bring about significant changes in the way future societies will function. Emerging technological environment allows for processes of information and knowledge sharing across the interfaces of government and civil society, the voicing of opinions, the expression of needs, and the strengthening of participation in ways that can enhance participatory democracy. In this new setting, we have access to a large amount of information about what is people's situation, what they think and what they believe.

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Policy makers, but also civil society groups, will have access to a huge amount of information that they should be able to use to elaborate policies, to take decisions, or to define rules of behaviours in a way that is well informed and legitimized by its fairness and transparency [5]. However, the utilization of this data appears problematic if we consider the complex, fragmented nature of this information, and its huge quantity. Policy-makers need feedback about their initiatives in order to align public policies with emerging civil society needs, requirements and expectations, while civil society is in need of transparency in the policy making process. One of the main challenges in this respect is that public policies have to operate in complex shifting environments structured by several intervening factors related to forces of change at global, EU, national, and regional levels. ICT can support policy-makers in modeling their policy design and implementation by capturing, analysing and acting upon a better assessment of their constituents' needs and expectations.

In this context there are several questions emerge about the optimal utilization of ICT for policy modeling [6]: how can we avoid being overwhelmed by this flood of information knowing that the quantity of this information will only increase in the future? How can we guarantee that some of the most relevant information will not be ignored, such as that originating from the lowest social strata of the population, or groups that are the most marginal to society? How can we make the best use of this information, and in particular, how should it be presented in a way that will be used to inform the policy elaboration process in a more targeted way? How can we manage the interaction and coordination across civil society agents exploiting this data, and in particular, how can we help people interact and coordinate in a way that is effective? How can we support knowledge sharing in a way that will not be too time consuming and avoid information overload? How can we extract the collective intelligence of different stakeholders, and in particular orient it so as to augment our ability to identify trends, and find solutions? How can we assure that this flow of information reaches the right government agencies or decision makers? With this regard, it is essential to be sure that the right information reaches the right (i.e. competent) body, institution or person.

According to the above detailed questions, one of the most important outcomes of our research is to explore and monitor the mid-term impacts of policy decisions. Impact can be captured through the "voice of citizens" by an online environment and serves as an additional input for fine-tuning decisions through the modeling. Feedback collected can be pre-processed with the data and text mining tools, in order to filter and aggregate stakeholders' opinion for the modeling and to modify model variables if it is needed. Finally, by using advanced complex event processing methods, our approach enables continual monitoring of a policy impact, providing a useful mechanism for managing the risk in policy implementation, especially in changing environments.

The approach applied will consist of the extension and operationalization of theories often referred to under the concept of the "attention economy" [7]. In particular, such theoretical approaches will be further extended to deal with information that originates from civil society agent interaction and participation, and more generally from agents involved in social and collective processes. We refer to this theory as social attention management.

The typical target audience for suggested policy modeling support is the local government. On this level of Public Administration (PA), the regulatory activities – the outcome of which is the policy – are complex enough to confront different ideas, approaches and opinions. At the same time they are close enough to the citizens in order to articulate location and context dependent arguments expressed by the citizens and other stakeholders. Due to the fact of the wide variety of local governments' size, culture, organization etc., the outcome of the project can be applied for a big variety of problems. Finally, local PA is especially vulnerable to failing in the implementation due to a very open information sharing space that alleviates opinion diversity. The paper will be structured as follows:

First, the challenges of the ICT utilization for policy modeling are detailed. Next theoretical background of policy modeling is discussed, followed by research overview. The proposed solution – policy modeling cycle and the corresponding system is presented in the following section. Finally, conclusion and future work are shown.

2. THEORETICAL BACKGROUND - POLICY MODELING AND GOVERNANCE

Involvement of citizens and businesses in policy-making processes is an increasingly important and a currently addressed issue. As a key enabling factor, policy modeling implies the application of methods and tools from a broad range of disciplines and integrates various lines of research and the state–of- the-art ICT in modeling complexity).

This multidisciplinary research area integrates the expertise of ICT and eGovernment experts, but also policy modellers, social and behavioural scientists with a common interest in modeling social dynamics. Policy modeling aims at developing ICT tools and applications allowing the modeling and simulations of policy options by the government, by an integrated and inclusive policy approach that involves all relevant state / public sector and civil society actors in the design, implementation and evaluation of policy initiatives. Recently there has been a range of significant technical and methodological advances in policy modeling analysis and the development of theoretical and policy simulation models.

Policy modeling is the method of developing a precise model that can help understand and evaluate available policy options in an actual situation. These models are necessary to bridge the gap between the complexity of the environmental/sociological/economic/political situations that exist and the decisions and understanding of the policy advisor that has to face these. Such modeling is typically based on statistical, econometric, systems dynamics, micro-simulation and agent-based simulations.

2.1 Policy Modeling Approaches

Models are simplified abstractions of the complex reality, developed for representation, analysis, or simulation. Modelling is the creative process of identifying and formalizing the underlying principles to formulate a model. Several methods are available for modelling the behaviour of a complex system, like policy making. The following approaches are relevant in policy modeling:

- Behavioural modeling [15]: it is used to understand the behaviour of complex systems over time. Computer simulations that are based on these models facilitate studying the behaviour of systems and the impact of alternative policies, using "if-then" relations and rules. In policy modeling behavioural models are relevant as they predict how people will likely react if the policy becomes implemented. There are social, economic and environmental models.
- System dynamics [17]: In system dynamics target system, including its properties and dynamics, is depicted by a system of equations which draw the future state of the target system from its actual state. System dynamics presents the macro level, it models the target system as an undifferentiated whole.
- Multi-level and micro-simulation models [10]: multi-level simulation models consist of at least two levels where the lower levels subordinated to the macro-level (i.e. households and national economy). Micro-simulation models represent the individual level with a number of attributes and a number of transition probabilities.
- Queuing models or Discrete Event models [3]: discrete event modeling has a long tradition in a wide variety of sciences. It represents a system, in terms of its entities and their attributes, sets, events, activities, and delays.
- Cellular automata [19]: cellular automata are often described as the spatial counterpart of non-linear differential equations. Cellular automata are useful mathematical tools for describing natural and social processes. The theory of cellular automata is immensely rich, with simple rules and structures being capable of producing a great variety of unexpected behaviours.
- Agent-based social simulation [8]: multi-agent modeling is based on a set of automata (called agents) for social simulation that is somewhat more complex in the internal processing (i.e. behaviour). Agents can be classified as self-contained programs that can control their own actions based on how they perceive their operating environment. Agents continuously monitor their environment and learn from it creating agent models while the simulation runs. Multi-agent systems (MAS) are composed of distributed heterogeneous units, where each agent manages its own activities on the basis of its local state and the information/messages received from other agents.
- Theory of complexity [12]: the approach of complexity is an interdisciplinary approach in science that can be used to explain highly complex phenomena in terms of simple rules. Although the elements of a system are

relatively simple, nonlinearity in their interactions may lead to highly complex dynamic behaviour.

Policy making or modeling tools are starting to be used across governmental and non-governmental organisations, to support policy making also in developing countries, and used extensively in health, education, criminal justice environment, urban planning, transport etc.

Most recent approaches to policy analysis should also be taken into consideration. In the 1990s several researches focused on the architectural issues of policy systems, emphasizing the importance of enforcement. [11], [13]. Marriott [13] modelled the policy life-cycle as editing, distributing and deleting policies. Avitable [1] developed an extended lifecycle approach, and differentiated a development phase (refinement, deploy, distribution, test), and an operational phase (activation/deactivation, enforcement, removal). The common characteristic of these approaches is that they are focusing on technological issues. A policy lifecycle model for system management was developed by Zhang et al [20], concentrating on internal organisational policies, and policy enforcement, identifying elements of the lifecycle as management objectives, policy definitions, policy deployment and policy enforcement. The active technological support for development and enforcement is vital during the lifecycle of a policy.

There are different approaches of evaluating policy impact mainly vary according to the specific policy to be evaluated, as well as according to the moment in which they are carried out: ex ante, monitoring and ex post. Ex ante analysis is a "what if" analysis, meant to capture differences between the proposed reform(s) and the status quo. Monitoring analysis is a "what's happening" process, meant to collect feedback while a new measure is deployed. Finally, ex post analysis, seeks for results achieved, given the initially fixed objectives. The most common evaluation methods include cost-benefit analysis (CBA), behaviour models, and techniques.

Another key issue in policy making processes is risk assessment. NAO [14] suggests six key questions which public administrations might ask themselves to assess whether they have a sound approach to managing risk. This is particularly important where initiatives require coordination between a number of parts of the same organisations or with other organisations. Key questions policy makers should consider in assessing their approach to risk management include: Do senior management support and promote risk management? Do the public administration support well thought through risk taking and innovation? Are risk management policies and the benefits of effective risk management clearly communicated to staff? Is risk management fully embedded in the public administration's management processes? Is the management of risk closely linked to the achievement of the public administration's key objectives? Are the risks associated with working with other organisations assessed and managed? A possible solution for preventing risks to undermine public policies deployed are Early Warning Systems [8], which showed their potential benefit also in policy modeling. Early Warning Systems bases on one or more models of how the phenomenon monitored behaves. The model is being used to show what is likely to happen next. These models can range from simple to very complex systems. Early Warning Systems can base on quantitative and qualitative approaches using for instance either a more formal forecasting oriented approach through e.g. simulations or a more informal foresight approach using e.g. scenarios.

Involvement of citizens/stakeholders in the policy making process has vital importance. This involvement is starting to take place through innovative approaches, based on direct participation of stakeholders, often convoyed through IT means. Edelenbos [9] defines interactive decision making as a way of conducting policies whereby a government involves its citizens, social organizations, enterprises, and other stakeholders in the early stages of the policy-making process. The difference with more traditional public policy procedures is that parties are truly involved in the development of policy proposals, whereas in classic opportunities of public comment, citizen and interest group involvement only occurred once the policy proposal had been developed. Obviously active involvement of stakeholders brings along also a series of problems, because in most cases it is quite different from traditional decision-making procedures, so separate organisational provisions have to be developed in order to conform to these innovative decision-making procedures. Evaluating the connection of this new policy practice with existing decision making and the elaboration of guidance supporting this new practice is definitively important.

2.2 Policy Risk Modeling and Management

Risk has become a prominent feature of our society [4] although human societies have always lived with risk [16]. Changes in legislation, politics, and public attitudes have produced an environment in which there is now much greater awareness of risk. As a result, both public and business organizations now give greater priority to risk management.

Risk is defined as this uncertainty of outcome, whether positive opportunity or negative threat, of actions and events. The risk has to be assessed in respect of the combination of the likelihood of something happening, and the impact which arises if it does actually happen. Risk management includes identifying and assessing risks (the „inherent risks") and then responding to them.

Risk relates to the chance of injury, damage, or loss of human, financial, physical, and natural resources under the care of a PA. Risk management seeks to conserve and to protect public resources from accidental loss. It proactively mitigates governmental risk through awareness, avoidance, assessment, containment, funding, and management oversight [2].

Common risk management processes as: a) determination the scope of investigation, b) risks identification, c) risks analysis and evaluation, d) risks communication and treatment and e) risks monitoring can be applied for policy risk management only in a limited sense. Public administration has a very low risk taking attitude; therefore the well-known steps of risk management have fairly different interpretation.

The opportunity of using tools to manage risks is particularly relevant for government organisations due to the extent of their effects on the stakeholders which, in many cases, comprise the social fabric of certain country: in this sense, the risk management activities generates improvements in the "quality of society".

In spite of the huge literature of the risk management, relatively few papers are discussing risk modeling and management specialties in public administration [18]. This research will target this questions as well, namely we will investigate the nature of risks in PA, the special characteristics of risks modeling and management. Based on the general suggestion of project management literature and best practices we should pay a special attention to the risk management issues in connection with policy modeling. More precisely, we will investigate methods for continuous risk monitoring that will enable proactive reaction in the case of problems.

3. The Proposed Policy Modeling Solution

The vision of our research is to address the aforementioned needs by specifying, developing and deploying a holistic framework and supporting tools for policy modeling. The framework includes a methodological concept, a procedure and a toolbox that will be sufficiently flexible to adapt to changing environments and to reflect civil society needs, taking into account potential legal barriers and enablers.

The overarching objective of the project is to investigate, develop and validate tools that can be used to explore the potential impact of policies by using risks identification and assessment and by exploiting all the information originating from the "crowd" of the citizens. In particular, our solution is concerned with:

- the identification, processing, and management of information available from the "state/ public sector - civil society information space", so that such information contributes to the better (in terms of democratic participation and ethics) and more effective (in terms of allocation of resource) regulation of the governance process.

- developing and validating new meta-model for policy monitoring that will be used for monitoring and exploring the risk in the policy implementation
- developing and validating an innovative approach for predictive monitoring of the key policy risk indicators (KPI) that will support policy attention management, i.e. awareness creation that a risk has appeared
- developing and validating a novel visualization metaphor for observing policy impact, that can be used for analyzing risk development.

In this sense, our research aims at addressing the needs of different stakeholders such as policy makers, and civil society agents interested in participating in a process of good governance.

Risk has become a decisive feature of our society and of public administration activity as well. Changes are discontinuous showing increased complexity and turbulence (e.g. Stuttgart 21 in Germany or the red mud tragedy in Devecser in Hungary). As a consequence, in both cases public organizations now have to give greater priority to risk management, and look for tools and techniques, which enable them to know more about their decisions' impact. In the proposed solution the policy maker is supported to become risk-aware through capturing, highlighting and mitigation the possible impact. Policies have to be regularly updated, because of the frequent changes in the legislation, economic and political situation, and appearing more often extreme environmental events (e.g. storms, flooding, related industrial accidents /like red mud/, volcanic ash, tsunami,). The policy making cycle starts with the articulation of the need for a new policy or policy update (figure 1). Policy maker has to identify the complexity of the problem area, to formalize the issue into a decision model, to collect data including the opinion of citizens and other stakeholders, and to explore the risks. Evaluating the complex model the consequences of several policy alternatives are concluded, based on which the policy maker can select the "quasi"-optimal solution. Figure 1 provides a big picture about the phases of risk aware policy modeling cycle, which is detailed below.

Apart from the policy modeling aspect, the proposed solution has an ICT-related added value. The impact will be captured not only through plain model calculations but in the "dimension" offered by interpretation of the "voice of citizens". The social computing element is part of the online environment and serves as an additional input for modeling. Feedback will be pre-processed with the data- and text-mining tools, in order to filter and aggregate stakeholders' opinion and if it is necessary modify the model variables. Our approach introducing a significant enhancement of the state of the art in following research areas:

- Utilization event-based feature of process modeling during the policy making process (e.g. policy maker is notified and alerted when certain policy has to be updated (e.g. environmental or financial disaster))
- Prediction of policy impacts through event based non-classical economic modeling, simulation
- Visualisation and monitoring of risks and impact of policies
- Policy modeling based on simulated behaviour
- Developing, analyzing, and packaging high quality, dynamic feedback based on active monitoring of social channels
- Agents-based decision support process.
- Exploitation of public sector's collective knowledge resources through semantic data and text mining
- Application of web 2.0 techniques, like social networking to support citizens' inclusion in policy making process (possibility of capturing citizens' opinion and voting)
- Developing, analyzing, and packaging high quality dynamic feedback models, which are constructed graphically or applying augmented visualisation.

It has potential to influence ever important, heavy problems in policy implementation (e.g. Stuttgart 21 in Germany) caused by the fact that social-cultural trends influence a lot policy making and modeling process and that societal aspects are not emphasized enough during policy modeling, resulting in failing to measure additional potential impacts of the proposed policy measures (risk).

3.1 Policy Modeling Cycle

In this section we provide a more detailed view of the main steps (1-5) of the policy modeling cycle of the proposed solution (Figure 1 gives a high-level overview about it):

3.1.1 Step 1 - Policy Maker Starts the Process and Customizes the Policy

This step starts with the problem perception. In some cases it is trivial since "the situation is given" (e.g. Iceland ash), but in other cases the problem is implicit, i.e. it has a hidden or tacit nature. Policy maker has to put the problem into context, highlighting the policy making aspects and providing the initial risks set in the context of the policy problem under investigation. It is done based on the policy attributes, which are set up in this step. The system offers (automatically or semi-automatically) an initial set of risks which have to be modelled.

Customization includes the following activities:

- Domain definition from a predefined list of domains (policies belongs to certain domains, like urban planning, energy, education, etc.)
- Type is selected from a predefined list of problem types (like energy consumption, social support in education)
- Community selection (action has impact to a given community, stakeholders)
- Indicator assignment (policy maker defines the initial set of parameters on which the impact of action will be measured, and describes the launching set of indicators to be used) - the "enactment" of the model should help the policy maker to recognize the relevant parameters and indicators
- Measurement means how the impact is measured through the indicators; the result of the measurement is fed back to the policy maker.

ICT elements applied at this step: ontology-based (domain, type, action, risks and community) development of the framework component.

3.1.2 *Step 2. Policy Maker Selects the Model*

In step 2 the system suggests to the policy maker the problem related model type from the model repository (e.g. forecasting model category to estimate the costs of certain policy alternative). In the next step the concrete models are selected, these models are used to model risks and impacts.

Step 2 covers the following activities:

- Domain model selection (domain model is determined by the problem domain of the policy, like e.g. energy, social support)
- Model type definition (these models belong to IT, like simulation or data mining models). During this step public sector's collective knowledge resources/public cloud are utilized to determine which model type/model would be the most appropriate for the investigated situation.

ICT elements: Model selection is supported by agents (according to the policy attributes, models are offered by agents).

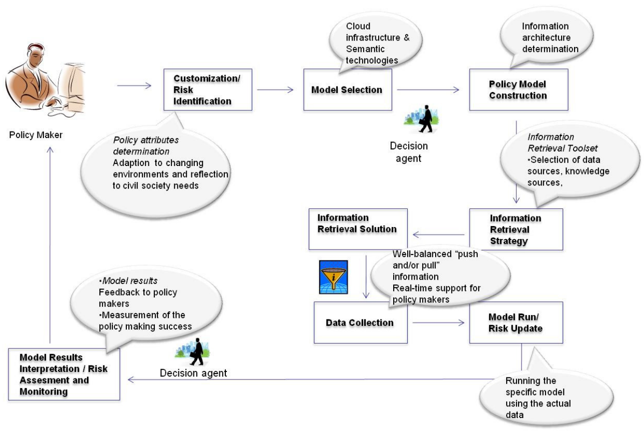

Figure 1. Policy Modeling Cycle

3.1.3 *Step 3. Policy Maker Constructs the Model*

This step is about the detailed model definition, parameters setting, based on the previously defined model. The system guides the policy maker in model setting, with predefined options suggestion and help. Policy maker has to give additional information, like the time scale (e.g. which time period is used in modeling, the location(s), or which variables are interesting and important.

Model construction covers the following activities:

- Selected models have to be specified in the context of the customization
- The concrete problem has to be generalized or mapped to the model context
- In parallel, the abstract model has to be customized
- Model should be used to collect citizens' opinion.

3.1.4 *Step 4. Policy Maker Selects and Applies Information Retrieval Strategy*

After model setting, policy maker has to define the information sources and the strategy to collect data including notification, alerting and analysis services which are foreseen. The system will offer an information set with several sources, like web sources (e.g. forums, blogs from public administration and other sites), social networks (like Facebook), public information sources, statistical databases (e.g. economic indicators). Based on the information source selection the information retrieval strategy is offered by the system. The information retrieval strategy also covers the requirements of the reporting structure (analysis services, detailed in the next step). One of the key challenges at this step is the data quality. We can't assume that all data which is required is available in the right format, and able to be integrated with other datasets. In order to avoid GIGO (garbage in/garbage out), we plan to apply a careful analyses of all data and their sources. Data cleansing will support us during data collection to reach the proper data quality.

3.1.5 *Step 5. Running the Model and Interpretation of the Results*

Running the model means, that

- the predefined data is collected from multiple sources using the proper toolset (e.g. crawlers)
- a special focus is given to the citizens' opinion about the planned policy measure, i.e. appropriate information is collected and processed.

The model running can be a one-step or cyclical activity - depending on the problems' character. Cyclical running is necessary, for example, when the system's alert or notification service is activated. The systems early warning capability is provided by event-driven policy attention monitor (PAM) component. This early warning capability is especially important in urgent situations, when the system is capable to alert and "advice" the policy maker to carry out certain predefined steps.

Interpretation comes after running the model. The tasks in this step are separated in two main groups:

a) The initial risks set defined in the first step of the cycle. Initial risks set have to be updated, because new risks can occur or some previously defined risk won't be relevant. Modeling output is utilized in risks assessment. The interpretation component of the system contains a risks assessment part, which helps the policy maker in risks prioritization and in planning the risk mitigation.

b) Policy maker in the problem statement phase (first step of the cycle), assigned (technically it is managed by the PAM) a set of indicators, in order to characterise, measure, operationalize the problem solving. In the interpretation phase, these indicators and their internal relations are used to evaluate the model output and draw possible conclusions (in most of the cases firing rules).

c) Possible impact of the policy alternatives will be shown to the policy makers through dashboards using visualisation and business analytics - beside of the textual summary of the conclusions - that are drawn from model output.

3.2 System Architecture

3.2.1 The System Layers

The system (framework and tools) will support the cooperation between policy makers and the community at least on three levels: (a) technical level, (b) semantic level and (c) organisational level - as presented in Figure 2. The system will manage the technical facilities of collaboration, which means, that policy makers and the community members will reach it and have a possibility to link and integrate it into their own IT environment. Another precondition of the cooperation between policy makers and the community is the semantic interoperability. Semantic interoperability means that the system will be based on a conceptual framework, a semantic layer, which continuously recreates a common conceptual background. Mainly the ontologies offering the map of cross-referential links among the various fields and "pieces" of mutually required knowledge may enable mutual understanding.

The conceptual framework is an outcome of the continuously (re-) emerging consensus between the stakeholders which also enhances their commitment. Finally we will investigate the organisational dimension of the fit of system. The proposed system has a strong effect on policy makers' organisation; business (administration) processes are to be reorganised, new skills are required, the way of policy making has to be changed. It interplays also with legal standards and requirements. We will analyse these factors and their inter-connections, in order to enhance the smooth integration of the system to policy makers' IT environment. From community side, it means that we apply intelligent techniques (like social analyses) to explore more information about the characteristics of the community (in terms of age, geographical location, professional background, motivation, etc.) while also taking into account and protecting intellectual and privacy rights.

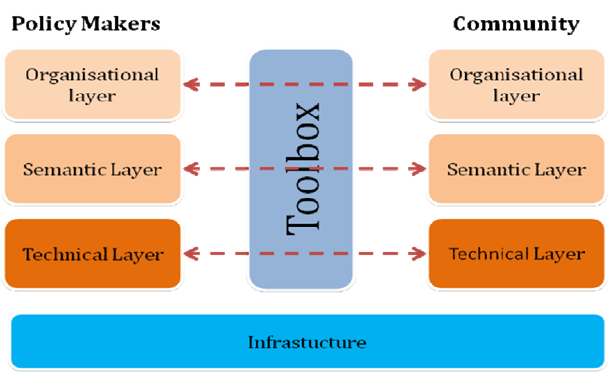

Figure 2 The System and its Environment

3.2.2 System Components

Due to the complex nature of the system architecture, it has to co-operate with a lot of information sources which are implemented using a variety of architectural patterns. Most of these co-operations are very dynamic in its nature, since the information sources can appear, disappear, change the way of communication, etc. The system will manage these complex, dynamic technical facilities of collaboration, what makes possible that both policy makers and the community can reach it and have a possibility to use it from their daily working IT environment. The cooperation with heterogeneous information sources raises typical interoperability problems. Moreover, the architecture has to be flexible enough to cope not only with the well-defined interfaces of e-government systems, but also with rapidly changing external systems (e.g. external databases, Web services) or un-structured and non-managed Web2.0 components such as forums, blogs, wikis, etc.

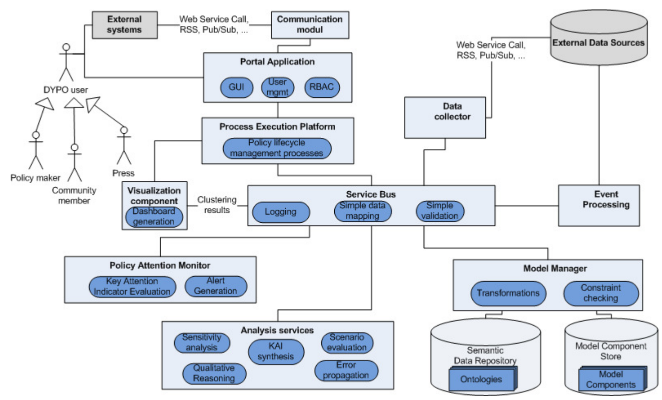

Figure 3 System Architecture

The cooperation between policy makers and the community is based on the semantic interoperability; the system has a semantic layer, which generates the common background of understanding. This will be facilitated by an ontology-driven development support. It will consist of a flexible framework which can be adapted to several domains (environment protection, city management, transport engineering, healthcare, etc.). In order to satisfy these application scenarios, the system requirements mandate a more dynamic, proactive policy monitoring and management environment with the following components:

Analytics. Policy makers have to be supported by multi-aspect analysis, results of which are easy to interpret and help the quick decision making. This means that the analytics must be embedded in the policy making process or delivered in-line to the end user applications.

Policy monitoring. Policy monitoring must help users to measure the efficiency of policies and the impact of environmental interactions on the success factors of a policy. Monitoring should support policy modification and give input for forecasting and scenario analysis by providing realistic data.

Configurability. Policy analysis and monitoring technology must be configurable to individual's needs. It should be self-service enabled so that the users can change the metrics or thresholds as well as the policy making rules on the fly. The presentation technology must be seamlessly embeddable to attention management applications, dashboards, portals, etc.

Event-driven operation. Dynamic policy management relies on the quick processing and interpretation of constantly changing information which is hard to implement using traditional technologies (incl. data warehouse). The framework must serve the increasing number and types of applications as well as support the varying service level demands from instant responses for short, relatively simple queries to slower responses to longer running, more complex ones. The environment will therefore use event-driven technologies, such as data consolidation and propagation to register and analyze events considering their significance. Inline analytics will automatically compare events against standard and user-defined thresholds and will generate alerts or alarms to the appropriate user groups in a timely manner.

Easy integration. Finally, policy implementers need technological solutions that can speed up information delivery and data presentation to end users at the appropriate points in their decision-making cycle. For this, they should look to flexible solutions, like a service-oriented architecture (SOA) or Web2.0 technologies to present analysis results at appropriate points in a flexible way.

Figure 3 presents the high level picture of the suggested system architecture, fulfilling the above requirements.

The flexible integration of external data sources (databases, complex information systems, web pages, unstructured Web2.0 components) is supported both by data collection in design time for policy creation and runtime event collection to provide input for the component evaluating the implementation of policies (Policy Attention Monitor.) These rely on a number of external data sources and filter and aggregate information to give useful input to modeling, analysis and runtime policy monitoring. Complex event processing technologies should be used to build a scalable architecture which can cope with thousands of events coming from distributed information sources.

Policy creation and lifecycle management is controlled by a central process execution environment, which interacts with other components via service invocations. As an underlying technology, we envisage the usage of XML Web services. Technical issues of service integration (logging, data transformation, request validation, etc.) will be managed by a service bus, following an architectural pattern widely used both in the industry and in e-government systems.

The system will use a set of ontologies to describe data in a semantically precise way. Ontologies will be derived from regulations (laws, regulatory frameworks, etc.), domain specific models and standards (e.g. logistics) and user-defined models (mindmaps, concept maps, etc). Moreover, the system will offer a set of initial domain specific languages to capture important properties affecting the success of a particular policy. For instance, the change in a local regulation on car usage can imply changes in traffic, taxes, and environmental conditions or even in the estate market. Modeling and evaluation of these implications needs domain metamodels, including well-formedness criteria, constraints (e.g. what is a valid traffic regulation) and presentation information.

The framework provides analysis services to support the early evaluation of policy impact and changes in the environment. As the underlying model will be based on a common ontology, these services can rely on powerful qualitative reasoning techniques as a common execution platform. Such techniques allow proving statements or searching for data by using abstraction instead of handling concrete values of performance simulation. Main analysis features include risk analysis (based on simple evaluations), sensitivity analysis (investigating the significance of parameter changes), error propagation analysis (estimating the effects of unplanned environmental, social or economical changes) and evaluation of scenarios (where scenario generation can be supported by the underlying model management framework). These services can be supported by the results of data clustering used by the visualization services and by qualitative markings produced by the user. This will obviously result in a higher level of abstraction and over-approximation, but still it can provide early analysis results which can be later refined on the basis of real measurements and events collected. An example can be the case study of transport policies, where the exact number of cars parking within a particular area is out of interest, instead, the range and the trend can be a useful input. On the other hand, analysis can also help monitoring by giving suggestions for indicators (KAI synthesis). In the parking example, if analysis results show that number of unauthorized parking permissions depends on the parking fee and frequency of checking, then number of permission requests will be candidate indicator for measuring the impact of a particular policy.

One key innovative feature of the solution is the monitoring of policy implementation and execution, in order to help policy makers by early warnings to adapt their policies to meet the required “performance”, captured by Key Attention Indicators (KAI). KAIs represent an equivalent to the Key Performance Indicators (KPI) in BPM context and constrain the search space for relevant changes. Policy Attention Monitor reacts on the complex events taking place in the environment and generates alerts which will be sent to the policy making process. This feed-back of "performance" indication will help to create a design loop for policies and will help to refine policies in order to reach the desired effects efficiently.

Trust evaluation based on external metrics for information sources can determine the importance of various incoming signals about events and reduce the “measurement noise”.

The system relies on the semantic integration of data captured by different taxonomies. This will be facilitated by a Model Manager component. This can be considered as a framework for model integration, including transformations and constraint checking on instance models. Meta-level information is stored in the Semantic Data Repository while model instance components (e.g. policy fragments, indicator templates) are preserved in a Model Component Store. Model components are packages in the system, containing not only the model itself, but the meta-information about their application as well, such as interpretation and presentation suggestions. Components can be replaced, expanded according to the user’s needs. These models can be applied by the policy makers, who will use the policy modeling process manager in policy modeling cycle. Policy modeling cycle steps are detailed in section 4.1.

Users can also access the system via a Portal GUI which offers a browser-based client with model patterns and policy suggestions for the support of design and dashboards for presenting the results of analysis and monitoring.

The portal application also performs user management and role-based access control functionalities. This will rely on the integration with existing user databases (e.g. community directories), however, a special care will be given to privacy and anonymity. An example can be the commenting of a municipal regulation: here the system could ensure that only affected users can comment the regulation, but will not reveal their identity for the policy maker, therefore enabling a more open discussion. This will be facilitated by built-in “anonymization” mechanisms. The architecture has to be flexible enough to cope with the needs of different organizations of policy makers: government offices, ministries, local authorities, municipal governments, etc. These organizations typically require different policy templates, policy lifecycle steps, data structures, or even the data sources/interfaces can be customized. The framework will use pre-defined, extensible model libraries (e.g., process model templates) and configurations (e.g., data schemas and suggestions to external services to cooperate with) to offer an easily customizable tool for different policy makers.

The overall framework will rely on widely used standards for Web-based system integration. Interfaces will be published in the form of XML Web services. Requirements on the usage of system interfaces (e.g. security, reliability of the offered service) will be also described as policies and will rely on WS-* standards. The system will offer Web2.0 interfaces (RSS, mashups) for its analysis and attention monitoring services. Data structures of the system will be described in a platform-independent way using XML schemas while actual system models will be stored in industry-standard technologies (XMI, EMF, etc.).

4. CONCLUSION

In this paper we proposed an ICT solution for policy modeling which emphasizes policy impact exploration, monitoring and risk management. The research has been started few months ago by an international consortium and now the planning and design phase is finished. During the planning phase we investigated several other ongoing and finished researches in the field of ICT support of policy modelling, like Ubipol project (FP7-ICT-2009-4: Activity – ICT 4.7.3 – ICT for Governance and Policy Modelling), which offer new governance platform and cooperation model for policy makers.

One of our solution's cornerstones is the deep understanding of the policy making process as a means to rationalize the role of individuals, groups and communities in its proceedings. In an attempt to offer improved empowerment (EC eGovernment Action Plan/Priority 1) and engagement of individuals, groups and communities, the system will on the one hand explore social policy to determine the type of model required, and on the other hand identify all the players involved in a policy modeling process. Nevertheless, activation and endorsement of social and civic engagement requires more than understanding the behaviours of the individuals. The actual problem lies in the unpredictability of collective behaviours. The system is encompassing both individual and collective participation in policy making processes by offering a platform of framework technologies and concepts intended to cope with the complexity of social interactions and environment.

Through its lifecycle approach, it turns individuals, groups and communities into active stakeholders of the policy making process. The proposed policy modeling cycle prescribes appropriate measures and instruments for the engagement of the community both in terms of defining the policy problem and eliciting and patterning its impact. The system is facilitating this process by using community enabling technologies.

Mapping of policy making to policy modeling results in process transparency and requires cross-domain matching. At this step policy modeling is enhanced with the application of semantic technologies, e.g. ontology will be used for a conceptual framework of the policy domain and cross domain interpretation. Complex event processing part of our platform (PAM) is innovative itself but it has an innovation potential through not only smaller events – sometimes looking unstructured – are recognized and signalled but also the relation among them can lead to predict consequences initiate measurements in a timely manner. Interpretation component will ensure the policy maker to understand the problem state, the several of proposed solutions in the full multidimensional context. This is only a necessary, but not sufficient condition; the public administrational culture shift is out of scope of this project. What is more, that the feedback is bidirectional cyclical, and timely. Not only policy makers get feedback from the citizens but in the course of application of the solution a full exchange of information takes place among the stakeholders. It ensure continuous improvement both of the quality of the policy making process and the policy itself, through the application of intelligent technologies, by enabling consistent change propagation while enhancing policy maker's productivity.

It empowers policy makers, by providing efficient feedback from their constituencies and communities. Feedback collected is planned to be pre-processed with the data and text mining tools, in order to filter and aggregate stakeholders' opinion about the modeling and to modify model variables if it is needed. Community feedback provides information not only to policy makers but it is connected on data level to the policy model through advanced technological solutions, like opinion mining and many other techniques, which are based on processing internet-based discussion forums, blogs, and comments. It covers entities, scope and technological elements, which should be matched to each other. The overall approach is the policy life-cycle model: plan-act-review-correct. The impact prediction is based on one hand the problem specific model results, on the other hand the articulated opinion, feelings of one or several communities having a fairly diverse values and multicultural attributes.

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6. REFERENCES

[1] Avitable, M. 1998, An Examination of Requirements for Metapolicies in Policy-Based Management, Munich Technical University, Thesis

[2] Baker, L. 2008. Risk Management. In Encyclopedia of Public Administration and Public Policy, Second Edition, ed. Jack Rabin; T. Aaron Wachhaus; Middletown, Pennsylvania, U.S.A., Taylor & Francis Group

[3] Banks, J., Carson, J., Nelson, B. and Nicol, D. 2005. *Discrete-event system simulation* - fourth edition. Pearson.

[4] Beck, U. 1992: Risk Society: Towards a New Modernity. London: Sage Publications.

[5] Bhatnagar, S. ; Sullivan, J. 2009. A Trans-Atlantic Analysis of Digital Media's Impact on Shaping Policy, http://www.edelman.com/Capital_Staffers_Index.pdf, 2009

[6] CROSSROAD, 2010. CROSSROAD, A participative Roadmap for ICT Research in Electronic Governance and Policy *Modeling*, ICT-2009.7.3 ICT for Governance and Policy Modeling, FP7-ICT-2009-4 Support Action (SA) Project, Deliverable D1.2, 2010

[7] Davenport, T. H.; Beck, J. C. 2001. The Attention Economy: Understanding the New Currency of Business. *Harvard Business School Press*. ISBN 1-57851-441.

[8] Davidsson, P. 2002. Agent Based Social Simulation: A Computer Science View. *Journal of Artificial Societies and Social Simulation 5 (1).* http://jasss.soc.surrey.ac.uk/5/1/7.html.

[9] Edelenbos, J. 2005. Managing stakeholder involvement in decision making: a comparative analysis of six interactive processes in the Netherlands, *Journal of Public Administration Research and Theory*, 2005.

[10] Hancock, R. and Sutherland, H. (1992) Microsimulation models for public policy analysis: new frontiers, London: STICERD, London School of Economics

[11] Koch, T., 1997. *Automated Management of Distributed Systems*, Berichte aus der Informatik. Shaker Verlag, 1997.

[12] Laszlo, E. 1996. *The Systems View of the World*. Hampton Press, NJ.

[13] Marriott, D.A.1997, *Policy Service for Distributed Systems*, PhD thesis, Imperial College of Science, Technology & Medicine, London, June 1997.

[14] NAO (National Audit Office) 2000: Supporting innovation: Managing risk in government departments. HC 864 Session 1999-2000. LONDON: The Stationery Office.

[15] Polderman, J.W. ; Willems, ; J.C. 1998. *Introduction to Mathematical Systems Theory: A Behavioral Approach*, 424 pages, Springer, New York.

[16] Rose, N. 2000. Genetic risk and the birth of the somatic individual, *Economy and Society*, Special Issue on configurations of risk (2000), 29 (4): 484-513. (with Carlos Novas).

[17] Sterman, John D. (2001). "System dynamics modeling: Tools for learning in a complex world". *California management review* 43 (4): 8–25.

[18] Wójcik, G. P. 2009. Management of risk in government organisations, *On-line Journal of First International Conference Quantitative Methods in Economics*

[19] Nowak, A., Maciej, L. 1996. *Modeling Social Change with Cellular Automata* in Ranier Hegselmann, Ulrich Mueller, and Klaus Troitzsch, eds., Modeling and Simulation in the Social Sciences from a Philosophy of Science Point of View. Amsterdam, Netherlands: Kluwer, 249–85

[20] Zhang Y, Liu X, Wang W, 2005. *Policy Lifecycle Model for Systems Management*, in: IT Professional, Volume 7 , Issue 2, Pages: 50 – 54, 2005